

Exam: *II partial - Reinforcement Learning*

Prof. Ruggero Carli and Prof. Gian Antonio Susto

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Provide complete and clear answers to the following questions. Clarity of the answers will be an important parameter in assessing your test.

Duration : 1 hour and half.

QUESTION 1 (4 points)

Consider a Markov decision process composed by three states, s_1, s_2, s_3 . Assume that

- at time $t = 1$ state s_2 is visited;
- at time $t = 2$ state s_3 is visited; and
- at time $t = 3$ state s_2 is visited.

Assume moreover that at time $t = 0$ the eligibility traces of all the states are initialized to 0, i.e., $E_0(s_1) = E_0(s_2) = E_0(s_3) = 0$. Let $\gamma = \lambda = 1/2$. Then describe the behavior of the eligibility traces for all the states at time $t = 1$, $t = 2$ and $t = 3$ (i.e., compute the values of $E_1(s_1), E_2(s_1), E_3(s_1), E_1(s_2), E_2(s_2), E_3(s_2)$ and $E_1(s_3), E_2(s_3), E_3(s_3)$).

QUESTION 2 (6 points) Check if each of the following statements is True or False, and provide a very short motivating/explanation sentence.

1. The advantage function is used as baseline within the REINFORCE algorithm.
2. Bootstrapping methods are typically characterized by smaller variance than Monte-Carlo methods.
3. In the *online updates* scenario, the Backward-view TD(λ) algorithm is equivalent to the Forward-view TD(λ) algorithm for any value of λ .
4. Let $\hat{v}(s, \mathbf{w})$ be an approximated function of $v_\pi(s)$. Then the TD target $R_{t+1} + \gamma\hat{v}(s_{t+1}, \mathbf{w}_t)$ is an unbiased estimate if $v_\pi(s_t)$.
5. Let $\hat{v}(s, \mathbf{w})$ be an approximated function of $v_\pi(s)$. Consider the Monte-Carlo updating step

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \alpha_t(G_t - \hat{v}(s_t, \mathbf{w}_t))\nabla_{\mathbf{w}}\hat{v}(s_t, \mathbf{w}_t).$$

Assume $\alpha_t = \frac{1}{t^2}$. Then $\mathbf{w}_t$ is guaranteed to converge to a local minimum of the mean-square error between $\hat{v}$ and v .

6. The updating step of the *one-step, action values, semi-gradient Expected Sarsa* algorithm (which is an off-policy method with function approximation) is given by

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \alpha\rho_t\delta_t\nabla\hat{q}(A_t, A_t, \mathbf{w}_t)$$

where

$$\rho_t = R_{t+1} + \gamma \sum_a \pi(a|S_{t+1})\hat{q}(S_{t+1}, a, \mathbf{w}_t) - \hat{q}(S_t, A_t, \mathbf{w}_t)$$

and ρ_t is the importance sampling ratio, that is, $\rho_t = \frac{\pi(A_t|S_t)}{b(A_t|S_t)}$.

QUESTION 3 (5 points)

Assume we want to represent points belonging to a portion of a two dimensional plane using tile coding, i.e., multiple tilings are used, each offset by a fraction of a tile width, see Figure 1 and Figure 2. We adopt two strategies both based on partitions given by uniform grids where features are squares. In particular:

- In the first strategy (Figure 1) we use three tilings where each tiling is composed by 16 tiles;
- In the second strategy (Figure 2) we use three tilings where each tiling is composed by 4 tiles.

Let $\mathbf{x}(s)$ be the binary features vector representing the state of a point (for instance the point depicted in both figures). Address the following questions:

1. What is the dimension of $\mathbf{x}(s)$ in the first strategy? and in the second strategy?
2. Is it possible to determine the number of elements equal to 1 in $\mathbf{x}(s)$ when the first strategy is adopted? and when the second strategy is adopted?
3. Which strategy provides generalization over a larger distance?

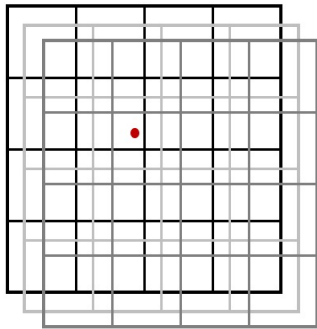


Figure 1: Tilings with 16 tiles

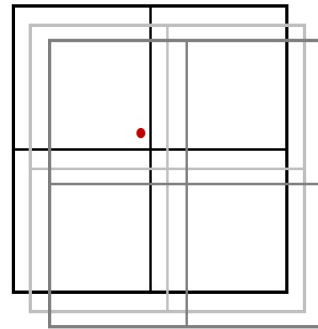


Figure 2: Tilings with 4 tiles

QUESTION 4 (10 points)

State the Policy gradient Theorem for the episodic case and show how it can be used to develop the REINFORCE algorithm.

QUESTION 5 (8 points)

Write the pseudo-code of the Episodic Semi-gradient Sarsa Algorithm for estimating $\hat{q} \approx q^*$.